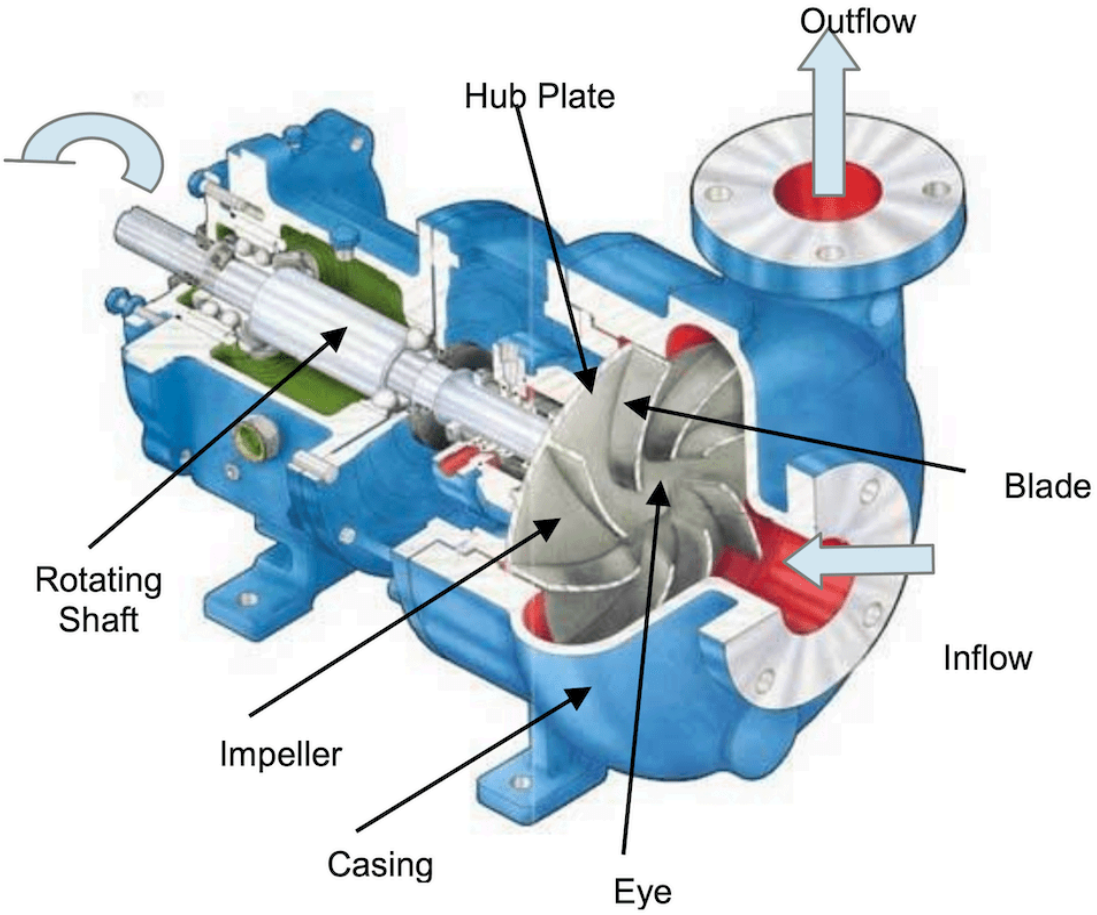
CENTRIFUGAL PUMP AND NPSH ANALYSIS:

A SIMULATION REPORT

Operating Point and Cavitation Assessment Using Excel and DWSIM.



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1 Introduction

A pump is a device used to transport fluid from one place to another. It works by pushing the fluid by the use of an impeller. One common example is the borehole pump in many homes which raises water from an underground reservoir to a storage tank. On a much larger scale, process plants such as oil and gas facilities use pipeline networks to move products between vessels. Pumps supply the energy that keeps the liquid flowing. Improperly selected or operated pumps can waste energy, deliver the wrong flow, or fail early, so designing and operating them correctly matters.

1.1 The Problem

- *What controls the flow a pump delivers on a given pipe system?*
- *How does a pump cavitate? When does a pump cavitate?*

To answer the following questions, we need to consider not just the pump but also the system in which it is being used. This report answers both questions for a typical water-transfer system, based on a manufacturer's pump curve.

1.2 Objectives and approach:

- ❖ To find the operating point of the pump on the pipe system.

- To show how a dirty pipe and a higher static head shift the operating point.
- To verify the Excel results in DWSIM.
- To assess cavitation risk with NPSHa vs NPSHr, including temperature.
- To identify and rank the failure modes (FMEA).

The approach was to build the system curve and a fitted pump curve in Excel, find where they meet, and check the result in a DWSIM simulation. Section 2 gives the theory, Section 3 describes the system and method, Section 4 presents the operating-point results, Section 5 checks them in DWSIM, Section 6 assesses cavitation, Section 7 gives the FMEA and Section 8 concludes.

2 Theory

Two curves decide where a pump will operate in a pipe system: the system curve and the pump curve. The operating point is where they meet.

2.1 System curve

The system curve reflects the head that a pipe system demands at varying flow levels. It is made up of two parts: static head and friction head. Static head, the difference in height between the two liquid levels, is unaffected by flow rate, but the friction head increases with flow:

$$H_{sys} = H_s + H_f (1)$$

The friction head is derived from the Darcy-Weisbach equation [1]:

$$H_f = f\left(\frac{L}{D}\right)\frac{v^2}{2g} \quad (2)$$

Velocity is related to flow via:

$$v = \frac{Q}{A} \quad (3)$$

Substituting equation (3) into equation (2) yields:

$$H_f = \left[\frac{fL}{2gDA^2}\right]Q^2 = kQ^2 \quad (4)$$

Where H_{sys} is the system head (m), H_s is the static head (m), H_f is the friction head (m), f is the friction factor, L is the pipe length (m), D is the pipe diameter (m), A is the pipe cross-sectional area (m²), g is acceleration due to gravity (m/s²), Q is the flow rate (m³/s), and k is the system friction constant. The flow rate Q is in m³/s but is calculated in m³/h in Excel, so a unit conversion is included within k

The constant k takes into account both the friction factor and the geometry of the pipe, given by the constant $k = \frac{fL}{2gDA^2}$. It is assumed to be constant because the friction factor, f, has very little dependence on the Reynolds number within fully turbulent flow, based on the Moody chart and the Colebrook equation. The Colebrook equation suggests that the value of f varies between approximately 0.01905 at 40 m³/h and 0.0177 at 96 m³/h, meaning that k is between about 2% below and 5% above its reference value throughout this range. At around the working flow, for which the evaluation of k was made, the error is less than 0.1% (Section 3.2). k is seen to increase with the length of the pipe, a smaller diameter, rough or dirty pipe walls, and with additional valves and fittings. Since

the head due to friction is proportional to Q^2, flowing at double the flow rate will result in a fourfold increase in friction head.

2.2 Pump curve

The pump curve shows the head that the pump is able to generate at various flow rates. The curve slopes downward as the flow rate increases. The manufacturer-supplied pump curve is available in the datasheet [2] and can also include efficiency data and the Net Positive Suction Head required (NPSHr).

Data points from the manufacturer's curve were plotted in Excel, and a third-order polynomial trendline was fit to the data:

$$H_{pump} = a_0 + a_1 Q + a_2 Q^2 + a_3 Q^3 (5)$$

Where H_{pump} is the pump head (m), Q is the flow rate (m³/h), and a_0 to a_3 are the fitted coefficients. a_0 is the head at zero flow, also known as the shutoff head. The pump curve is relatively flat near zero flow, then becomes steeper; fitting a third-order polynomial allows us to model this curvature allows this curvature to be modelled without chasing small reading errors, as a higher-order fit would.. The fitted equation is $H_{pump} = 15.6 - 0.00835Q + 0.00039Q^2 - 0.00000996Q^3$, with Q expressed in m³/h, and it overlaps with the points from the datasheet to within 0.07 m (the largest discrepancy occurs at 80 m³/h). The fit is not extrapolated beyond the datasheet flow range.

2.3 Operating point

The operating point is defined by the flow (Q_{op}) and head (H_{op}) at the intersection between the pump and system curves:

$$H_{pump}(Q) = H_{sys}(Q) (6)$$

In Excel, this was solved using Goal Seek. At flow rates less than Q_{op}, the pump head is greater than the head the system needs, speeding up the flow. At flow rates greater than Q_{op}, the pump head is less than the pipe head loss, slowing the flow down. This continues until the flow rate is equal to Q_{op}, at which point the head is the exact head delivered by the pump.

If the system curve rises because of a greater static head or a larger value of k due to pipe contamination, the intersection point will be higher on the pump curve, resulting in a lower flow rate and a higher head.

2.4 Pressure and Head

DWSIM gives pressures, but the pump and system curves are all in metres of head. To compare these graphs, it is necessary to convert pressure to head, and this conversion is given by:

$$H = \frac{\Delta P}{\rho g} (7)$$

Where H is the head (m), ΔP is the pressure difference (Pa), ρ is the density of the fluid (kg/m³), and g is the acceleration due to gravity (m/s²). For a pump, ΔP is the pressure difference between the outlet and inlet. Head is a distance and does not depend on the fluid, but the pressure needed to reach that head will depend on the fluid density. For water, a pressure increase of about 1 bar is roughly 10.2 m of head.

2.5 NPSH and Cavitation

Cavitation occurs when vapour bubbles formed in the low-pressure areas of the pump are crushed in the area of the impeller. It should be mentioned that Water boils not only when it is hot but whenever its pressure falls to its vapour pressure, which is why it boils at a lower temperature at high altitude. (such as at high elevations). At the pump inlet (which has the minimum pressure), the formation of vapour bubbles occurs when the pressure falls to the vapour pressure. The impeller then carries these vapour bubbles to a high-pressure area where they are violently destroyed in a process called cavitation, which is manifested by noise, vibration, impeller pitting, and a loss of flow and head [3].

Assessment of the potential for cavitation is carried out by examining two values, NPSHa and NPSHr, and comparing the two. NPSHa (Net Positive Suction Head Available) can be defined as the margin by which the inlet pressure is above the boiling point, in units of head, and is a function of the suction side of the system. NPSHr (Net Positive Suction Head Required) is defined as the head margin required to prevent cavitation; it depends on the pump, is published in the datasheet, and increases at higher flow rates. Typically, NPSHr is published by manufacturers as NPSH3 [3, 4], the NPSH at which the pump head decreases by 3% (due to cavitation). The criterion for the datasheet is not given, so for this paper, NPSHr is assumed to be NPSH3. (Note that cavitation may be present even before this point is reached, and a slightly higher NPSHa does not ensure a cavitation-free pump).

NPSHa is given by:

$$NPSHa = \frac{P_s - P_v}{\rho g} + z_s - H_{f,s} (8)$$

Where P_s is the absolute pressure at the liquid supply surface (Pa), P_v is the vapour pressure of the liquid at its current temperature (Pa), ρ is the liquid density (kg/m³), g is the gravitational acceleration (m/s²), z_s is the height of the liquid surface above the pump inlet (m, negative in the case of a suction lift), and $H_{f,s}$ is the suction pipe head loss due to friction.

Cavitation is expected if NPSHa is below NPSHr. Therefore, NPSHa must be greater than NPSHr for acceptable performance, which is expressed as:

$$NPSHa > NPSHr (9)$$

In practice, NPSHa should be higher than the pump's NPSHr by a certain margin, because NPSHr is established under test conditions [4] that may not exactly match the actual conditions of the system. In the present report, this margin is evaluated both as the NPSH margin ($NPSHa - NPSHr$) and the ratio of $NPSHa/NPSHr$. The necessary margin ratio depends on the application and the suction energy of the pump [4]. A ratio of 1.3 is used here as a screening criterion (Section 3.4).

Temperature affects NPSHa mainly through the vapour pressure.. As the liquid is heated, the vapour pressure will rapidly rise, decreasing the first term in equation 8 and lowering NPSHa. While the pipe, lift, and pump stay constant, the water is now much closer to boiling, shrinking the safety margin.

3 System and method

3.1 System description

The system is a water-transfer line made of a supply tank, a suction pipe, a pump and a discharge pipe (Figure 1). The pump is an HRZL IS100-80-125C horizontal centrifugal pump running at 2900 rpm. Its datasheet [2] gives a rated point of 80 m^3/h at 12.5 m head, with an efficiency of 64% and an NPSHr of 4.2 m.

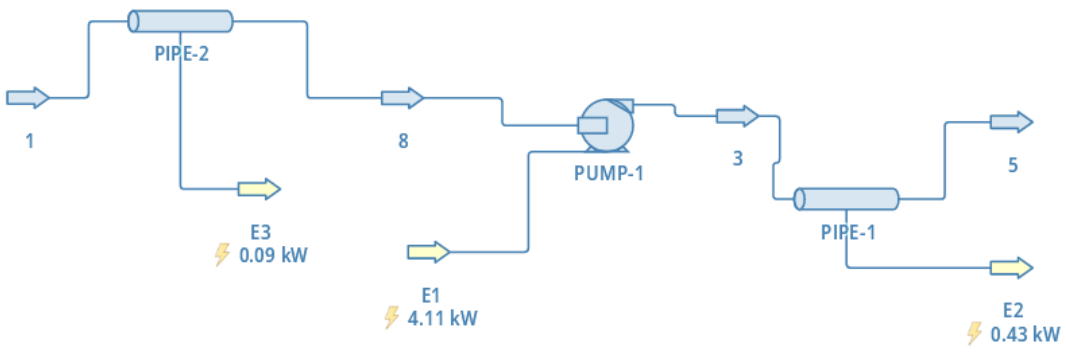


Figure 1: DWSIM flowsheet of the pump system.

Both pipes are raw steel with a 100 mm internal diameter (114 mm external) and a roughness of 4.57×10^{-5} m (Figure 2). The suction pipe is 5 m long and rises 2 m

from the tank to the pump. The discharge pipe is 90 m long and rises 6 m from the pump to the delivery point. The total static head is therefore 8 m. The supply and delivery points are at atmospheric pressure, and the base case uses water at 25 °C.

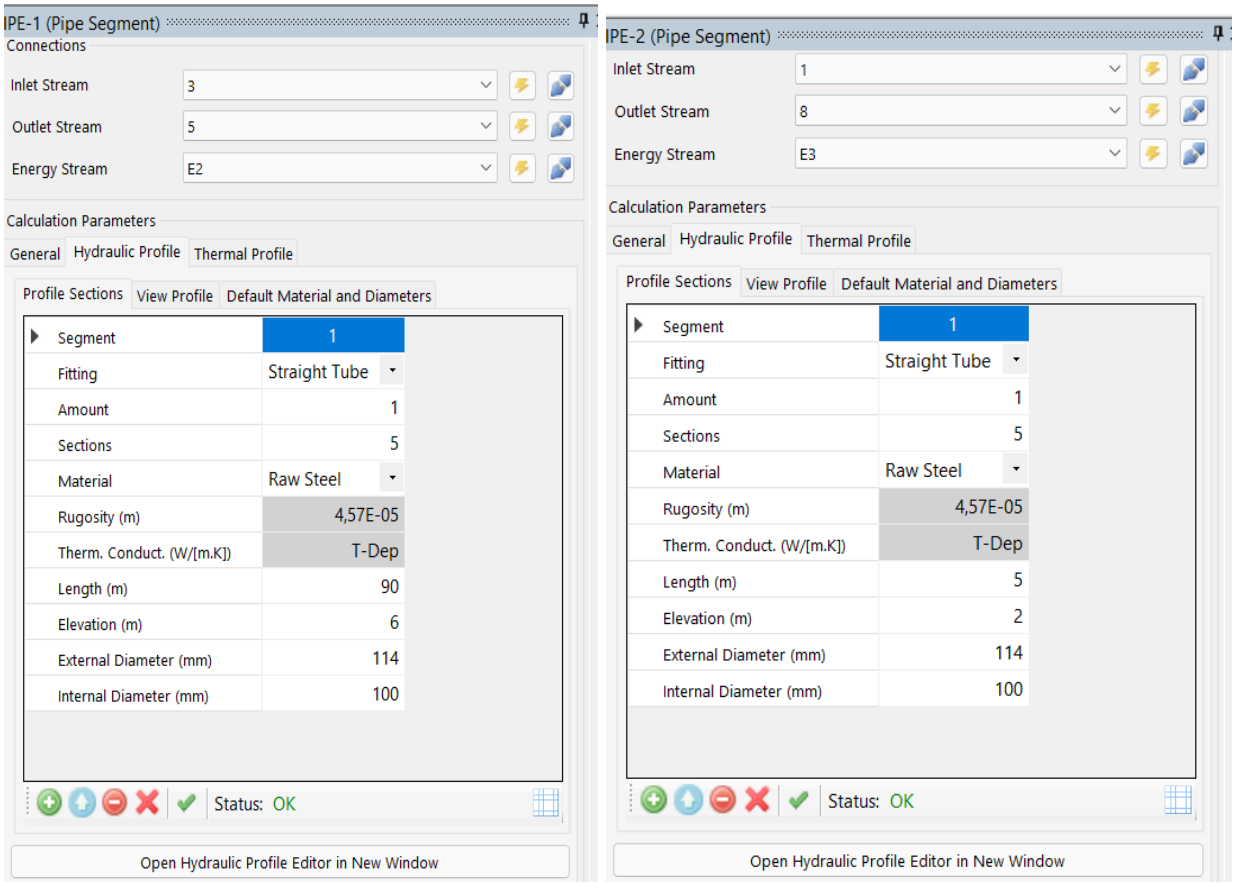


Figure 2: Pipe data in DWSIM. PIPE-1 (left) is the discharge pipe and PIPE-2 (right) is the suction pipe.

3.2 Excel model

The system curve was constructed from equations 1 to 4 with a static head of 8 m and a friction constant of $k = 0.001093467$ m/(m³/h)². The system head is given by $H_{sys} = 8 + 0.001093467Q^2$, where Q is the flow in m³/h. The pump curve was derived from a third-order polynomial fit of the datasheet points (equation 5). The operating point was found using equation 6 by setting the system curve (

H_{sys}) equal to the pump curve (H_{pump}) and solving it with Goal Seek. Two additional cases were run to explore the magnitude of the shift in the operating point:

- Higher static head: H_s was increased from 8 m to 9 m while keeping k constant.
- Dirty pipe: k was increased from 0.001093467 to 0.002 while keeping H_s constant.

The friction constant k includes all the system piping, made up of a 5 m suction line and a 90 m discharge line, for a total of 95 m of 100 mm diameter pipe. The constant k was evaluated using equation 4 at a reference flow rate of 71 m³/h, which is near the final operating flow rate. For this flow, the velocity is $v = Q/A = 2.51$ m/s (where $A = 0.007854$ m²) and the Reynolds number is $Re = \rho v D/\mu = 2.8 \times 10^5$, using $\rho = 997$ kg/m³ and $\mu = 0.891 \times 10^{-3}$ kg/(m·s) for water at 25 °C [6]. The flow is thus turbulent. The relative roughness is $\varepsilon/D = 4.57 \times 10^{-5}/0.1 = 4.6 \times 10^{-4}$. Using the Colebrook equation [1], $f = 0.0181$, and equation 4 gives $k = 0.001093$ m/(m³/h)² (converted from m³/s by dividing by 3600^2). Since f changes slowly with flow, for the final 70.6 m³/h operating flow rate, k would change by less than 0.1%. Away from the reference flow, the error increases: f is about 5% larger at 40 m³/h and about 2% smaller at 96 m³/h. This is a difference of about 0.2 m of friction head at 96 m³/h, and it is negligible at the operating point. For the suction line itself, $k_s = k \times 5/95 = 0.0000576$ m/(m³/h)², which is used in Section 3.4. For the dirty pipe case, $k = 0.002$, which corresponds to $f \approx 0.033$. This implies a wall roughness of about 0.6 mm, which is roughly 14 times that of the clean pipe, representing an encrusted or corroded pipe.

3.3 DWSIM Verification

This identical system was simulated in DWSIM [5] using the STEAM TABLES IAPWS-IF97 property package, with the two pipes above and the pump's head, efficiency, power, and NPSHr curves entered at 2900 rpm (Figure 3a). The operating flow rate from Excel (70.59 m³/h) was entered as the feed flow. If the Excel operating point is correct, the head supplied by the pump should exactly match the head needed by the pipes, allowing the water to arrive at the delivery point at atmospheric pressure. Pressures were then read at the pump inlet,

pump outlet, and delivery point, and the pump head was calculated via equation 7. The pump head, the delivery pressure, and the available NPSH were then compared to the results from the Excel calculations.

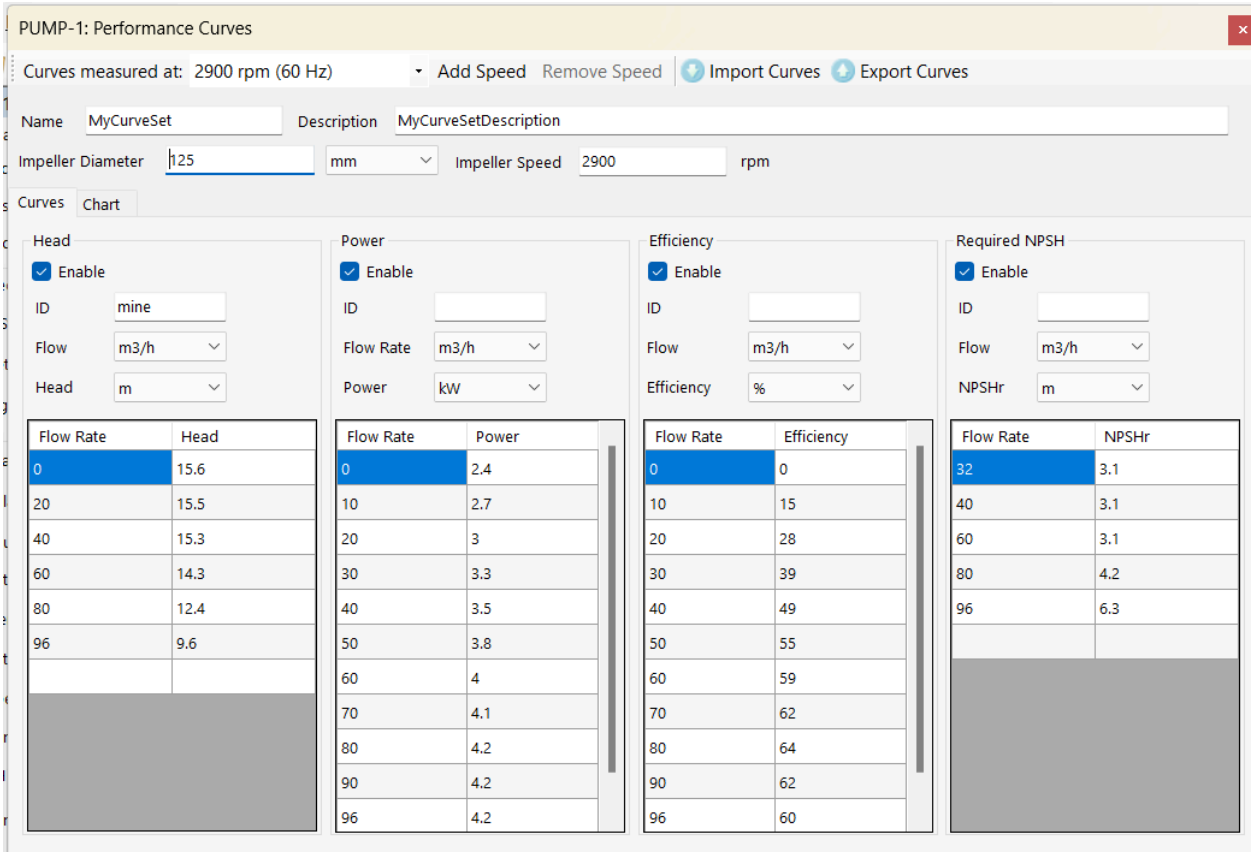


Figure 3a: Pump performance curves entered in DWSIM (data from [2]).

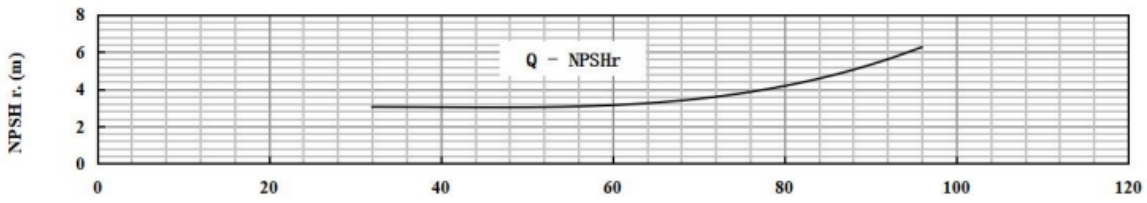


Figure 3b: Manufacturer NPSHr curve for the IS100-80-125C [2]. DWSIM uses the five points in Figure 3a. The Excel NPSHr table (Figure 9a) uses a finer

reading every 8 m³/h. Interpolating the five DWSIM points at 70.6 m³/h gives 3.68 m, within 0.03 m of the Excel value of 3.71 m.

3.4 Cavitation assessment

NPSHa was calculated in Excel from equation 8, with P_s = 101.3 kPa, z_s = −2 m (the pump sits 2 m above the supply level) and a suction friction constant of 0.00005755 m/(m³/h)². NPSHr was taken from the datasheet curve, and held constant at 3.1 m below 32 m³/h, where the datasheet curve starts. The check was made at the operating flow for water at 25, 60, 70, 75, 80 and 90 °C, using the vapour pressure and density at each temperature [6]: 3.2 kPa and 997 kg/m³ at 25 °C, 19.9 kPa and 983 kg/m³ at 60 °C, 31.2 kPa and 978 kg/m³ at 70 °C, 38.6 kPa and 975 kg/m³ at 75 °C, 47.4 kPa and 972 kg/m³ at 80 °C, and 70.1 kPa and 965 kg/m³ at 90 °C. NPSHr was read from the manufacturer's curve (Figure 3b) at 8 m³/h intervals (Figure 9a) and linearly interpolated at the operating flow. Between 64 and 72 m³/h (NPSHr = 3.3 m and 3.8 m): NPSHr = 3.3 + (70.6 − 64)/(72 − 64) × (3.8 − 3.3) = 3.71 m The curve is convex, so linear interpolation slightly overestimates NPSHr here, which is conservative. The points were read by eye, so NPSHr carries an uncertainty of about ±0.1 m. Each temperature was classed by the ratio NPSHa/NPSHr: Adequate margin if it is at least 1.3, Marginal if it is between 1.0 and 1.3, and Cavitation expected if it is below 1.0. The 1.0 limit follows from the NPSHr definition (Section 2.5). The 1.3 value is an assumed screening criterion, not a manufacturer limit or a requirement of [4]: required margins depend on the application [4]. Because it is an assumption, Table 1 (Section 6) shows how the temperature limit changes with it.

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4 Results: operating point

4.1 Base case

The pump curve and the system curve meet at 70.6 m³/h and ~13.45 m (Figure 4). Of this head, 8 m is static lift and 5.45 m is friction.

Flow	Pump curve	pipe line
0	15.6	8
8	15.55306048	8.069981871
16	15.52544384	8.279927486
24	15.48655296	8.629836843
32	15.40579072	9.119709944
40	15.25256	9.749546787
48	14.99626368	10.51934737
56	14.60630464	11.4291117
64	14.05208576	12.47883977
72	13.30300992	13.66853159
80	12.32848	14.99818715
88	11.09789888	16.46780645
96	9.58066944	18.07738949

flow	70.59584499
pump curv	13.44993275
pipe line	13.44959039
difference	0.00034236

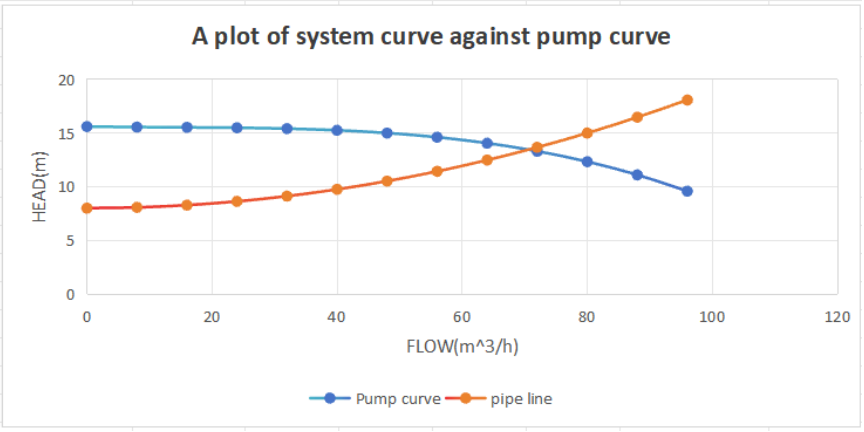


Figure 4: Pump curve and system curve for the base case.

The pump therefore runs to the left of its rated point (80 m³/h, 12.5 m). At 80 m³/h the pipe needs 15.00 m, but the pump can only give 12.33 m, so the flow settles lower, where the two match.

At 70.6m³/h the datasheet efficiency is about 62%, close to the pump's peak of about 64% near 80 m³/h, so the pump is running near its best efficiency point.

4.2 Higher static head

With the static head raised from 8 m to 9 m and k unchanged, the system curve moves up and the operating point slides up the pump curve (Figure 5). The flow drops from 70.6 to 66.52 m³/h, a fall of about 5.8%, and the head rises from 13.45 to 13.84 m.

Flow	system curve(9	pump curve		Flow guess	66.52005346
0	9	15.6		pump head	13.83859172
8	9.069981871	15.55306048		pipe head	13.83850014
16	9.279927486	15.52544384		difference	9.1583E-05
24	9.629836843	15.48655296			
32	10.11970994	15.40579072			
40	10.74954679	15.25256			
48	11.51934737	14.99626368			
56	12.4291117	14.60630464			
64	13.47883977	14.05208576			
72	14.66853159	13.30300992			
80	15.99818715	12.32848			
88	17.46780645	11.09789888			
96	19.07738949	9.58066944			

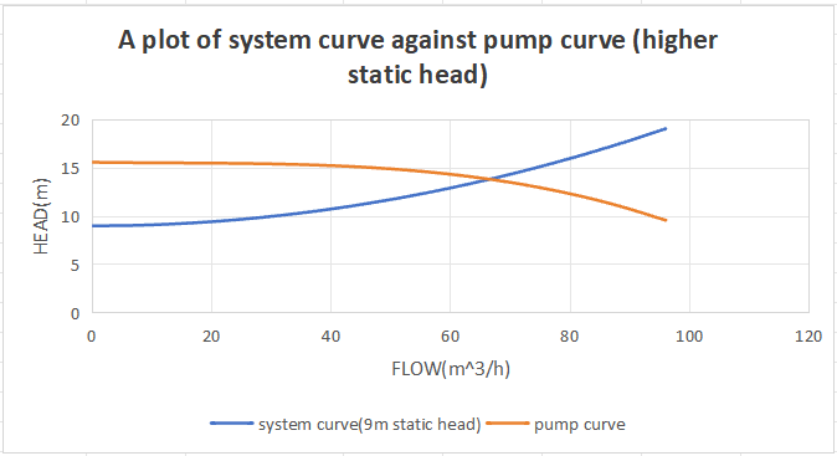


Figure 5: Operating point with a static head of 9 m.

4.3 Dirty pipe

With k raised from 0.001093467 to 0.002, the friction in the pipe nearly doubles and the system curve gets steeper (Figure 6). The flow drops from 70.59 to about 57.2 m^3/h, a fall of ~19%, and the head rises to about 14.5 m.

Flow	dirty pipe,	Pump curve
0	8	15.6
8	8.128	15.55306048
16	8.512	15.52544384
24	9.152	15.48655296
32	10.048	15.40579072
40	11.2	15.25256
48	12.608	14.99626368
56	14.272	14.60630464
64	16.192	14.05208576
72	18.368	13.30300992
80	20.8	12.32848
88	23.488	11.09789888
96	26.432	9.58066944

Flow guess	57.20192213
pump head	14.54408906
pipe head	14.54411979
difference	-3.07291E-05

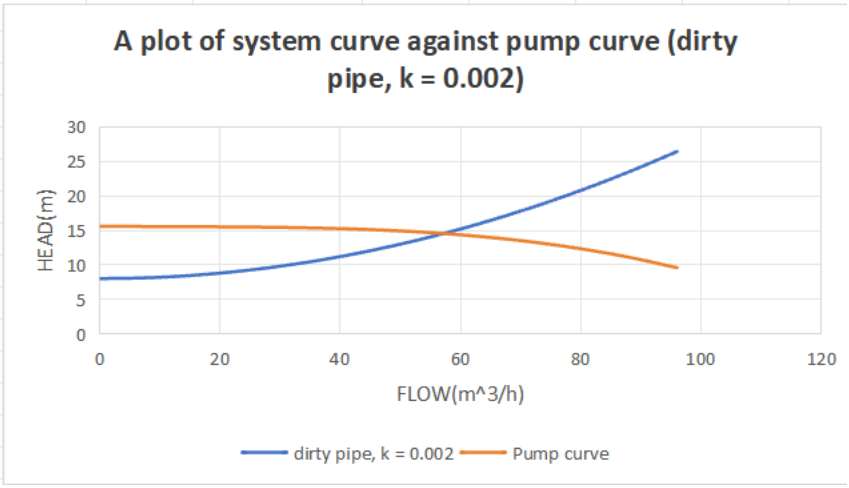


Figure 6: Operating point with a dirty pipe (k = 0.002).

Both cases agree with the theory in 2.3: the system curve moves up, so the operating point moves to a lower flow and a higher head. The dirty pipe has the bigger effect. Friction grows with Q^2, so it makes the curve steeper as the flow rises, while extra static head only lifts the whole curve.

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5 DWSIM verification

The Excel operating flow of 70.59 m^3/h was entered in DWSIM as a feed flow of 19.55 kg/s, which is 0.01961 m^3/s at the pump inlet. With this flow, the pressure at the delivery point (stream 5) comes out at 102,033 Pa, which is around 0.7 kPa above atmospheric pressure (101.3 kPa), or about 0.07 m of head. The pump head DWSIM gives is therefore the head the pipes need, to within 0.07 m (about 0.5% of the pump head), so the DWSIM model confirms the Excel operating

point. The small surplus means DWSIM's own balance point is about 0.3 m^3/h higher than Excel's.

The pump pressure rise is 132,376 Pa, from 78,967 Pa at the inlet to 211,343 Pa at the outlet (Figures 7 and 8). Using the stream density of about 997 kg/m^3 in equation 7, this gives a head of about 13.5 m, within 1% of the Excel value of 13.45 m.

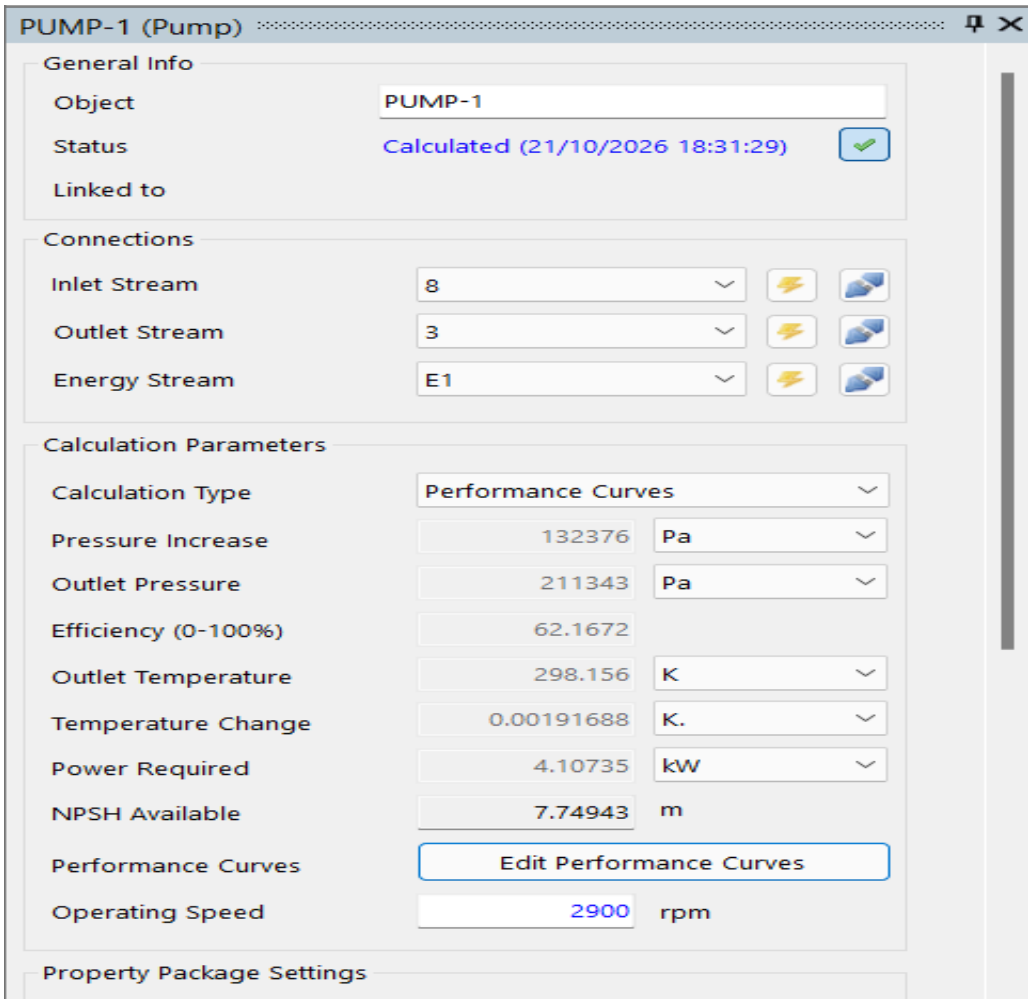
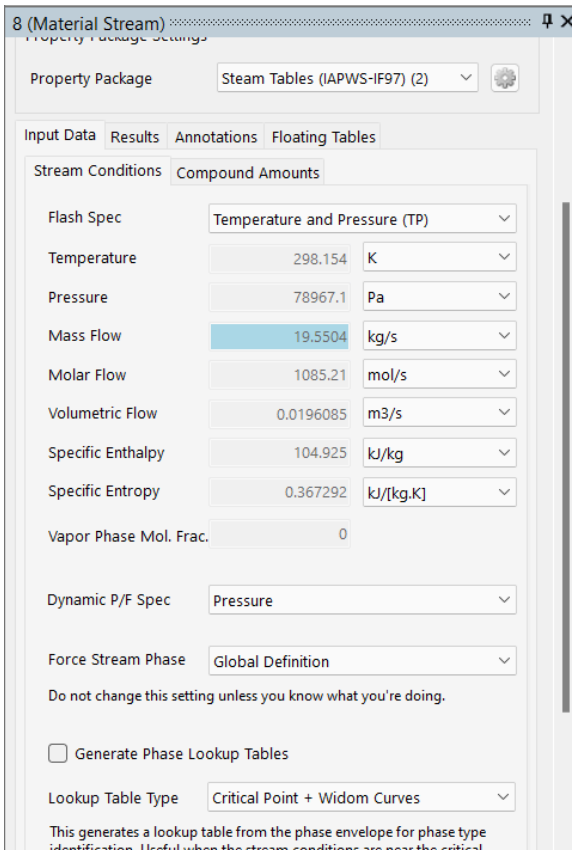
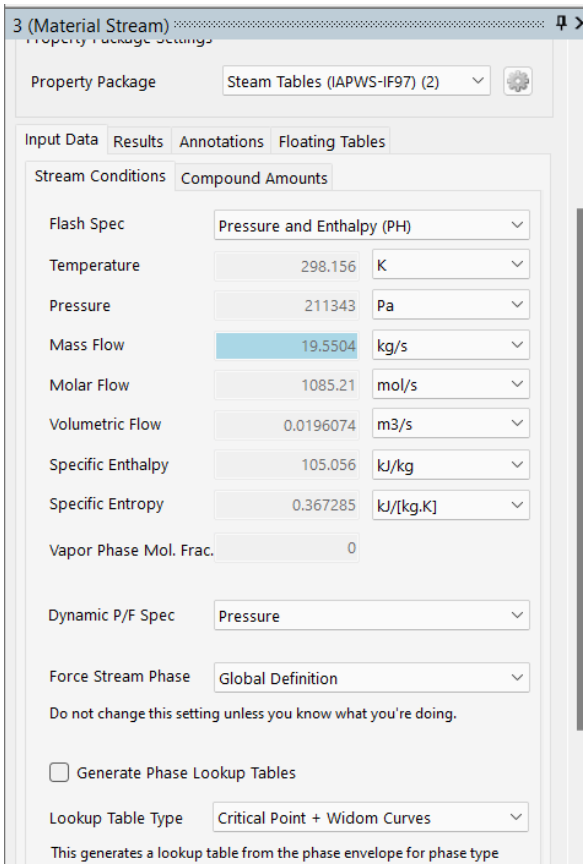


Figure 7: DWSIM pump results.


8 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
Input Data
Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
Temperature and Pressure (TP)
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K
Pressure
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Pa
Mass Flow
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kg/s
Molar Flow
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mol/s
Volumetric Flow
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m3/s
Specific Enthalpy
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kJ/kg
Specific Entropy
0.367292
kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
Lookup Table Type
Critical Point + Widom Curves
This generates a lookup table from the phase envelope for phase type


3 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
Input Data
Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
Pressure and Enthalpy (PH)
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K
Pressure
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Pa
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kg/s
Molar Flow
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mol/s
Volumetric Flow
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m3/s
Specific Enthalpy
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kJ/kg
Specific Entropy
0.367285
kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
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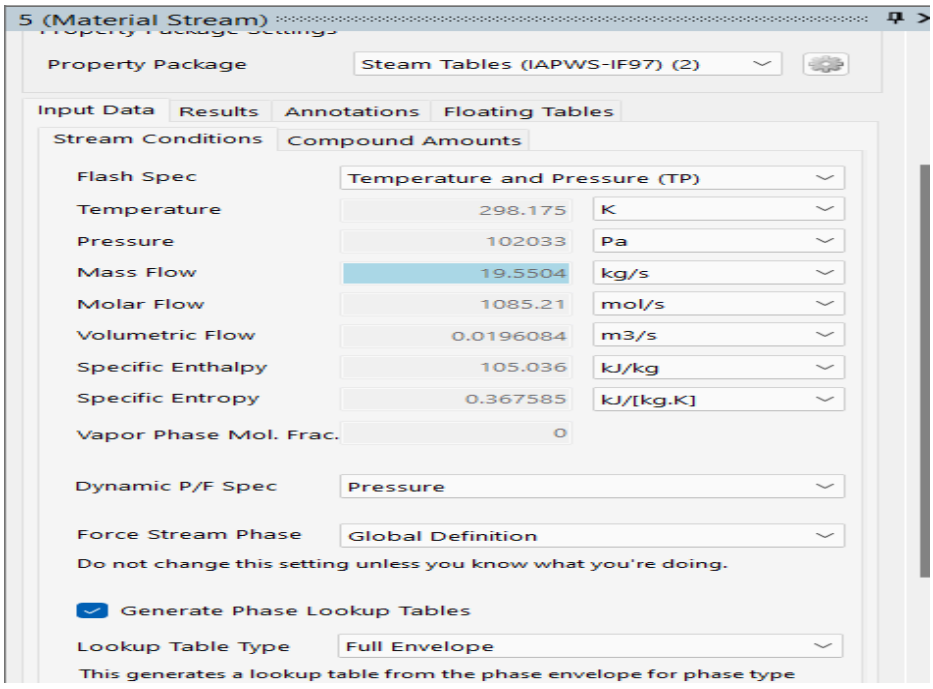

5 (Material Stream)
Property Package
Steam Tables (IAPWS-IF97) (2)
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Results
Annotations
Floating Tables
Stream Conditions
Compound Amounts
Flash Spec
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mol/s
Volumetric Flow
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kJ/kg
Specific Entropy
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kJ/[kg.K]
Vapor Phase Mol. Frac.
0
Dynamic P/F Spec
Pressure
Force Stream Phase
Global Definition
Do not change this setting unless you know what you're doing.
Generate Phase Lookup Tables
Lookup Table Type
Full Envelope
This generates a lookup table from the phase envelope for phase type

Figure 8: Pressures at the pump inlet (stream 8), pump outlet (stream 3) and delivery point (stream 5).

The pressures also match the elevations. On the suction side the pressure falls from 101.3 kPa to 79.0 kPa, about 2.3 m of head: the 2 m lift plus about 0.3 m of friction. On the discharge side it falls from 211.3 kPa to 102.03 kPa, about 11.2 m: the 6 m rise plus about 5.2 m of friction in the 90 m pipe. Together this is about 5.5 m of friction in DWSIM against 5.45 m in Excel.

The NPSH available in DWSIM is 7.749 m, against 7.746 m in Excel, a difference of 0.004 m, about 0.05%.

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6 NPSH and cavitation

At the operating flow of 70.6 m³/h, the NPSHr interpolated from the datasheet curve is 3.71 m (Section 3.4, Figure 9a) . NPSHa was calculated for water at six temperatures (Figure 9b).

Flow	Npsha	hf_suction	NPSHR	Margin	Ratio
0	8.032646359	0	3.1	4.932646359	2.591176245
8	8.028963103	0.0036833	3.1	4.928963103	2.589988098
16	8.017913333	0.014733	3.1	4.917913333	2.586423656
24	7.999497052	0.0331493	3.1	4.899497052	2.58048292
32	7.973714257	0.0589321	3.1	4.873714257	2.572165889
40	7.940564949	0.0920814	3.1	4.840564949	2.561472564
48	7.900049129	0.1325972	3.1	4.800049129	2.548402945
56	7.852166796	0.1804796	3.1	4.752166796	2.532957031
64	7.79691795	0.2357284	3.3	4.49691795	2.362702409
72	7.734302591	0.2983438	3.8	3.934302591	2.035342787
80	7.66432072	0.3683256	4.2	3.46432072	1.824838267
88	7.586972335	0.445674	5.3	2.286972335	1.431504214
96	7.502257438	0.5303889	6.3	1.202257438	1.190834514

Figure 9a: NPSHa, NPSHr, margin and ratio against flow for water at 25 °C (NPSHr read from Figure 3b at 8 m³/h intervals).

L	M	N	O	P	Q	R	S
TEMP	pv	rho	NPSHA	margin	ratio	verdict	
25	3.2	997	7.745873305	4.035873305	2.08783647	Adequate margin	
60	19.9	983	6.156974529	2.446974529	1.659561868	Adequate margin	
70	31.2	978	5.022345599	1.312345599	1.353731967	Adequate margin	
75	38.6	975	4.271161027	0.561161027	1.151256342	Marginal	
80	47.4	972	3.368516963	-0.341483037	0.907956055	Cavitation expected	
90	70.1	965	1.011648246	-2.698351754	0.272681468	Cavitation expected	

Figure 9b: NPSHa, margin and ratio at different water temperatures (P_v in kPa, ρ in kg/m³, , both from [6]; NPSHa and margin in m).

At 25 °C, NPSHa is 7.75 m, a margin of 4.04 m (ratio 2.09), so the margin is adequate. At 60 °C, NPSHa is 6.16 m (margin 2.45 m, ratio 1.66), still adequate, and at 70 °C it is 5.02 m (ratio 1.35). At 75 °C it is 4.27 m (ratio 1.15), which is marginal: above NPSHr, but below the 1.3 ratio used as the margin. At 80 °C it falls to 3.37 m, which is 0.34 m below NPSHr (ratio 0.91), so cavitation is expected. At 90 °C, NPSHa is only 1.01 m (ratio 0.27), far below what the pump needs.

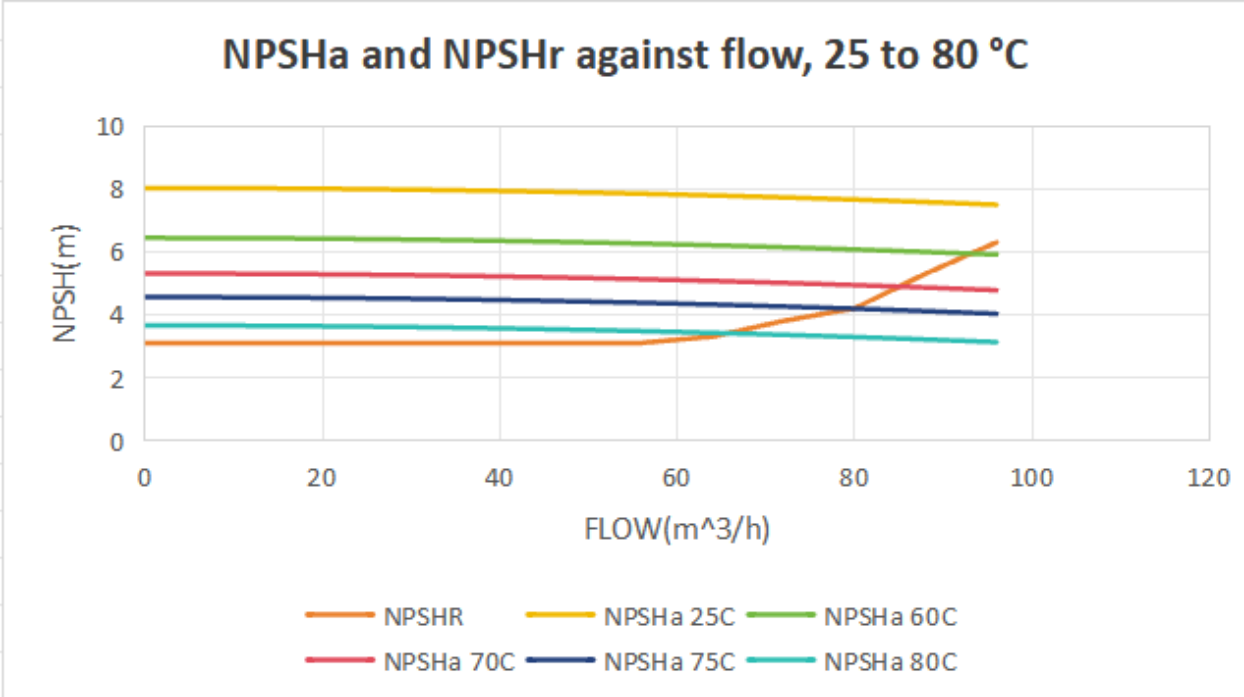


Figure 10: NPSHa at different temperatures against the NPSHr curve.

At 25 °C, NPSHa stays above NPSHr over the whole flow range. At 60 °C and 70 °C, the margin only disappears near the end of the curve, at about 93 and 85 m³/h. At 75 °C, NPSHa meets NPSHr at about 80 m³/h, so at the operating flow of 70.6 m³/h the pump is above NPSHr but with little margin to spare. At 80 °C, NPSHa crosses NPSHr at about 66 m³/h, so the operating flow of 70.6 m³/h is already past the crossing and cavitation is expected.

Table 1: Highest water temperature at the operating flow for different required margin ratios (NPSHr = 3.71 m; temperatures interpolated linearly between the tabulated values).

Required Ratio	**Max water Temperature**
1.0 (NPSHa = NPSHr)	about 78 °C
1.1	about 76 °C

1.3 (used here)	about 71 °C
1.5	about 65 °C

Whatever criterion is chosen, the pump has adequate margin at 25 and 60 °C and is at or beyond its limit between roughly 75 and 80 °C.

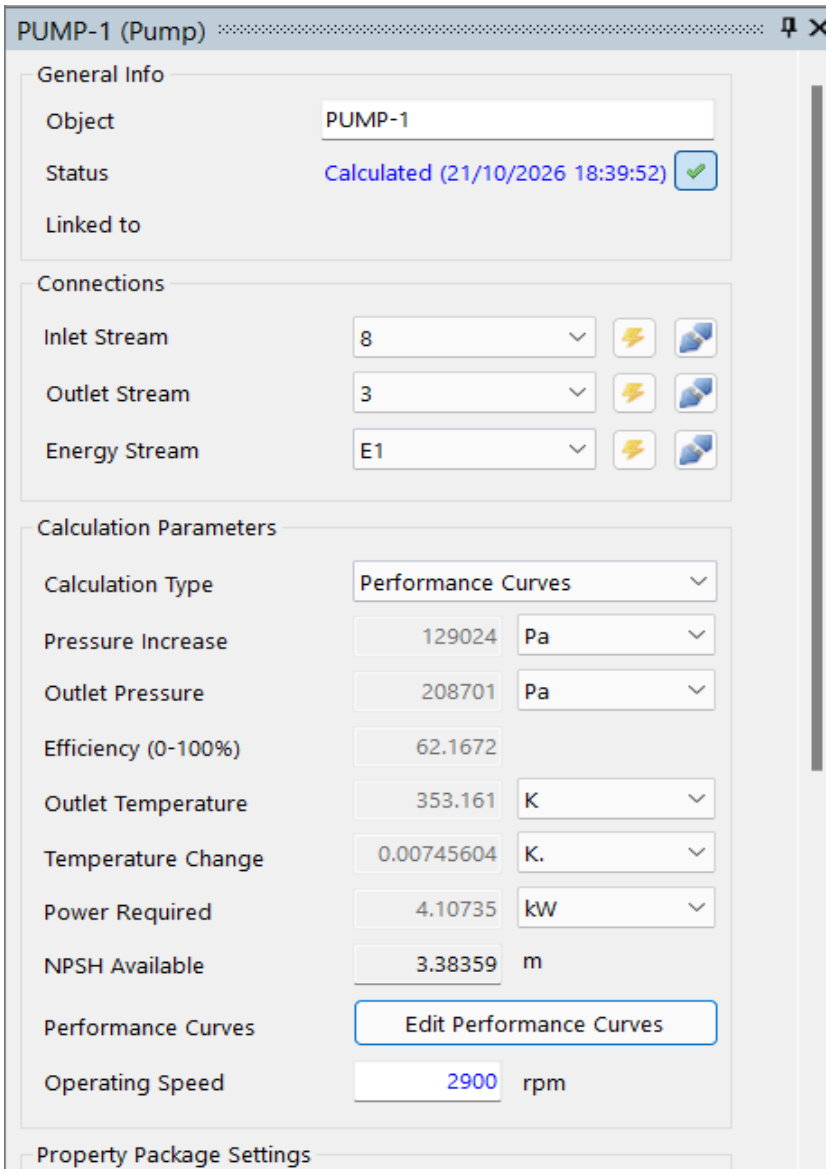


Figure 11: DWSIM NPSH available at 80 °C.

The DWSIM run at 80 °C was kept at the same volumetric flow of 70.59 m³/h, so the mass flow fell from 19.55 to 19.05 kg/s because the water is lighter. At 80 °C,

DWSIM gives an NPSH available of 3.38 m. This is close to the Excel value of 3.37 m (a difference of 0.015 m) and is about 0.3 m below the NPSHr of 3.71 m (ratio 0.91). DWSIM and Excel therefore agree that cavitation is expected at this temperature. The pump pressure rise falls from 132,376 Pa to 129,024 Pa because the hot water is lighter (972 kg/m³ against about 997 kg/m³ at 25 °C), but the head is still about 13.5 m, the same as at 25 °C, since the head comes from the pump curve and not from the liquid.

The cause is the vapour pressure. Going from 25 to 80 °C raises it from 3.2 to 47.4 kPa, which removes about 4.4 m from NPSHa. The pipe, the lift and the pump have not changed. The water is just closer to boiling. To avoid cavitation in this situation, cool the water, lower the pump or move it closer to the tank, or run at a lower flow.

7 Failure mode and effects analysis (FMEA)

7.1 Method

An FMEA lists the ways a system can fail, what each failure does and how serious it is [7]. This one covers only the failures the simulation can say something about, so bearing and seal failures are not included. Each failure mode is rated Low, Medium or High for severity (how bad the effect is) and likelihood (how likely the cause is), and Easy, Moderate or Hard for detectability (how easily it is noticed before damage is done). The ratings are engineering judgement, and a qualitative scale was used because no failure-rate data is available for this system. Priority is High when severity is High and likelihood is at least Medium. Within the same priority, modes backed by results from this report rank higher.

7.2 Results

Table 2: Failure modes and the evidence from this report

Failure mode	Cause and effect	Evidence	Control
Cavitation	NPSHa falls below NPSHr (hot water, suction lift, suction friction, high flow). Impeller pitting, noise, vibration, lost head [**3**]	At 25 °C, NPSHa is 7.75 m against NPSHr of 3.71 m (ratio 2.09). At 80 °C it is 3.37 m (ratio 0.91), so cavitation is expected. At the operating flow the limit is about 78 °C, or about 71 °C if a ratio of 1.3 (assumed) is required.	Limit water temperature, keep the suction pipe short, keep a margin, listen for noise and check vibration
Restricted suction pipe or strainer	Debris or scale raises the suction friction, so NPSHa falls and cavitation follows	The suction friction H_f,s lowers NPSHa one for one (equation 8). At 25 °C it would have to rise about 15 times (0.29 m to about 4.3 m) to reach NPSHr, and at 60 °C about 9 times (to about 2.7 m)	Suction pressure gauge, clean the strainer on a schedule

Loss of prime or dry running	Air enters the suction line or the water level drops, so the pump runs without water and overheats	Not modelled. The pump sits 2 m above the supply level, and a standard end-suction centrifugal pump is normally not self-priming (check the datasheet), so it depends on an airtight suction line staying full	Foot valve, airtight suction pipe, low-level cutoff
Fouled discharge pipe	Scale or corrosion raises friction, so flow falls	In the dirty-pipe case (k = 0.002), flow falls from 70.6 to 57.2 m^3/h and head rises from 13.45 to 14.5 m	Compare flow and head against the clean-pipe curve

Table 3: Ratings and priority

Rank	Failure mode	Severity	Likelihood	Detectability	Priority
1	Cavitation	High	Medium (High above about 78 °C)	Moderate: noise and vibration only appear once it has started	High

2	Restricted suction	High (leads to cavitation)	Medium	Moderate: needs a suction gauge	High
3	Loss of prime or dry running	High	Medium (the lift makes it more likely)	Moderate	High
4	Fouled discharge pipe	Medium	Medium	Easy: a drop in flow or head shows against the curve	Medium

7.3 Discussion

Cavitation ranks first because it can damage the impeller and bearings and starts quietly, and because Section 6 shows how close the pump is to it: adequate margin at 60 °C but cavitation expected at 80 °C. A restricted suction pipe ranks next because it attacks the same margin. Loss of prime is not modelled, but it is a real risk for a pump above the water level. The fouled pipe does no damage and is the easiest to spot, since the flow simply drifts down against the clean-pipe curve.

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8 Conclusion

1. The pump runs at 70.6 m^3/h and 13.45 m on the base system, to the left of its rated point (80 m^3/h). Of this head, 8 m is static lift and 5.45 m is friction. The efficiency at this flow is about 62%, near the pump's best.
2. A higher static head (9 m) moves the operating point to 66.52 m^3/h and 13.84 m, a flow drop of about 5.8%. A dirtier pipe (k = 0.002) moves it to about 57.2 m^3/h and 14.5 m, a drop of about 19%. Friction has the bigger effect because it grows with Q^2.
3. DWSIM confirms the Excel result. At 70.6 m^3/h the water reaches the delivery point within 0.07 m of head of atmospheric pressure , the pump head is about 13.5 m against 13.45 m, and the NPSH available is 7.749 m against 7.746 m.
4. At the operating flow the pump has adequate margin at 25, 60 and 70 °C (NPSHa/NPSHr of 2.09, 1.66 and 1.35), marginal margin at 75 °C (1.15), and cavitation is expected at 80 °C and 90 °C (0.91 and 0.27). The 1.0 limit comes from the NPSHr definition and the 1.3 screening ratio is an assumption, so the limiting temperature is about 78 °C at a ratio of 1.0 and about 71 °C at 1.3 (Table 1). The cause is the vapour pressure, which removes about 4.4 m of NPSHa between 25 and 80 °C. DWSIM agrees at 80 °C (3.38 m against 3.37 m in Excel, both below the NPSHr of 3.71 m).
5. The FMEA ranks cavitation first, followed by a restricted suction pipe and loss of prime (all high priority), with a fouled discharge pipe at medium priority. The first two are backed by the NPSH results in this report.

Limitations: The pump curve points were read by eye from the datasheet, k was treated as constant although the Colebrook friction factor varies from about 0.01905 at 40 m^3/h to 0.0177 at 96 m^3/h, and NPSHr was held at 3.1 m below 32 m^3/h. Excel and DWSIM use the same pump curve, so their agreement checks the pipe side of the model, not the pump data. The system model includes only straight-pipe friction: valves, fittings, the tank exit and the strainer are not

modelled, so real friction and k would be higher and the real operating flow lower. The velocity head at the delivery point (about 0.3 m at 2.5 m/s) is ignored. NPSHr was read by eye (about ±0.1 m), the 1.3 margin ratio is an assumed screening value, and the temperature limits are linear interpolations of a curved vapour-pressure relationship. The restricted-suction FMEA entry assumes the flow stays at the operating value. The datasheet NPSHr is assumed to be NPSH3.

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